MTH 137(B) Dr. James Ashe
Quiz 1.1-1.4 August 13, 2026 **Name:**

No credit will be given unless you show your work. This quiz will count as two homework grades.

Solve the following equations.

1. $4(x+4) = 6(x-2)$

2. $\dfrac{3x}{4} - 4 = 1 - \dfrac{7x}{8}$

3. $5|4x-5| - 3 = 9$

4. $\dfrac{1}{F} = \dfrac{1}{v} + \dfrac{1}{n}$ for n.

5. $B = \dfrac{1}{9}t(f-s)$ for f.

Solve the following equation; identify the equation as an identity, inconsistent, or conditional equation; and determine for which values the equation is undefined.

6. $\dfrac{9y}{y+1} = 7 - \dfrac{9}{y+1}$

Solve the following word problems

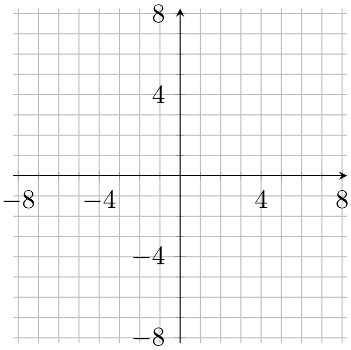
7. Farmer John is going to fence off a lot and then cross-fence to divide it into four smaller lots. If he has 516 feet of fencing, how much area will be fenced?

8. How many liters of a 70% alcohol solution must be mixed with a 50 liter solution that is 30% alcohol to get a mixture that is 50% alcohol.

9. James can grade a pile of 30 quizzes in 2 hours. It takes Catherine 1 hour to grade the same stack of quizzes. If they work together, how long will it take to grade the quizzes?

Perform the indicated operation(s).

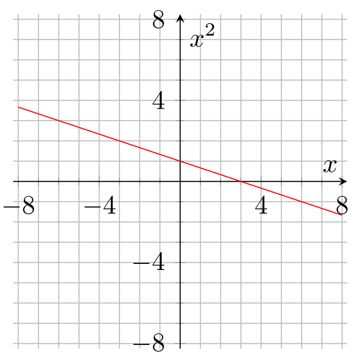
10. Find the center and the radius of the circle. Then graph the circle$(x+3)^2 + y^2 = 9$.



11. Write the standard equation for the circle centered at $(9, 4)$ and passing through the origin.

12. Find the slope, if it exists, of the line passing through the points $(5, 1)$ and $(-2, 1)$.

13. Write an equation for the line shown on the right.



14. Find an equation for the line which passes through the point $(5, -4)$ and is perpendicular to $y = \frac{1}{2}x + 4$.